IMO 2026 Route-Neutral Grading Schemes and Reference Solutions

Method-independent MathArena-style 7-point rubrics

Independent draft based on the reviewed problem and solution sources

18 July 2026

Status and scope. These are proposed, independently written grading schemes for review and automated-evaluation use. They are *not* official IMO coordination schemes. This route-neutral v4 includes the full reviewed reference solution for every problem and scores alternative proofs by mathematical function rather than by resemblance to the displayed reference route.

Route-neutral grading policy

Route-neutral grading policy. Grade the mathematical substance of the submission, not its similarity to the reference solution. A complete and correct proof receives 7 points regardless of method, notation, or auxiliary constructions. For an alternative proof, award each rubric item for fulfilling the same logical function described there; examples from the reference route are illustrative rather than mandatory. A downstream item is awarded only when its needed input has been proved or independently replaced by an equivalent result. Mere mention of a milestone does not earn proof credit, and a step depending on a material false claim does not earn dependent points.

General coordination principles

- Award an item only when the stated mathematical obligation is justified, not merely mentioned.
- A complete correct proof receives 7 points regardless of method or notation. Match alternative proofs to the logical function of each item; named objects from the reference solution are illustrative, not mandatory.
- Multi-point items state how their internal partial credit should be split.
- Later points may be awarded when independently correct, without double-counting the same work.
- A downstream item that logically depends on an unproved or false claim is not awarded unless the needed conclusion is independently re-established.
- A total of 7 requires a complete and internally coherent proof; accumulating listed phrases or conclusions cannot override a substantive gap or contradiction.
- A minor local algebraic or notational slip should affect only the rubric item it materially damages.

Contents

1 Problem 1

Problem statement.

There are 2026 integers greater than 1 written on a blackboard, not necessarily all different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces them with

$$\gcd(m, n) \qquad \text{and} \qquad \frac{\operatorname{lcm}(m, n)}{\gcd(m, n)}.$$

He continues to make moves while it is possible to do so.

(a) Prove that, regardless of Confucius's choices, after finitely many moves exactly one integer M on the blackboard is greater than 1.

(b) Prove that the value of M does not depend on Confucius's choices.

Core target. $\nu_p(M) = \gcd_{1\le i\le 2026} \nu_p(a_i)$ for every prime p.

Pts	Logical function	Criterion
2	**Establishing finite termination**	For proving that every move makes strict progress in a well-founded order, so an infinite play is impossible. The reference route uses the lexicographic pair consisting of the number of entries greater than 1 and the product of all entries, but any correct monovariant or finite-descent argument is accepted. Award 1 point for a valid monotone framework with all move types substantially analyzed but a minor gap in strictness or well-foundedness.
1	**Characterizing the terminal position**	For proving that a terminal board has exactly one entry greater than 1: two such entries would permit another move, while the process cannot eliminate every non-unit entry.
1	**Finding a choice-independent invariant**	For identifying, prime by prime, a quantity preserved by every move and capable of determining the final prime exponent. One example is $I_p = \gcd(\nu_p(t_1), \dots, \nu_p(t_{2026}))$; any equivalent invariant earns the point.
2	**Proving preservation of the invariant**	For rigorously proving that the proposed invariant is unchanged by one move and hence throughout the process. In the valuation route, the local update is $(x, y) \mapsto (x, y - x)$ for $x \le y$, and $\gcd(x, y - x) = \gcd(x, y)$. Award 1 point for the correct local transformation or an equivalent preservation calculation with the global invariance argument incomplete.
1	**Determining the final number**	For evaluating the invariant at the terminal board, obtaining $\nu_p(M) = \gcd_i \nu_p(a_i)$ for every prime p (or an equivalent complete characterization), and concluding that M is uniquely determined by the initial board.
7	**Total**	

Coordination note. Part (a) and part (b) are logically independent after termination is established. A correct alternative invariant or any rigorous well-founded monovariant receives the corresponding points.

Reviewed reference solution

Write $d = \gcd(m, n)$. The product of the two selected entries changes from mn to

$$d \cdot \frac{\operatorname{lcm}(m,n)}{d} = \operatorname{lcm}(m,n) = \frac{mn}{d}.$$

Let r be the number of entries greater than 1, and let P be the product of all entries. If $d = 1$, then P is unchanged while r decreases by one. If $d > 1$, then P decreases strictly and r cannot increase. Thus the ordered pair (r, P) decreases lexicographically after every move. Since lexicographic order on pairs of positive integers is well-founded, the process terminates.

Every legal move leaves at least one of its two outputs greater than 1. Consequently, the process cannot end with all entries equal to 1. At termination there cannot be two entries greater than 1, since then another move would be legal. This proves (a).

For (b), fix a prime p. If the current entries are $t_1, \ldots, t_{2026}$, consider

$$I_p = \gcd\bigl(\nu_p(t_1), \ldots, \nu_p(t_{2026})\bigr),$$

using the convention that the gcd of a list of zeros is 0. Suppose the selected entries have p-adic valuations $x \le y$. Their two new valuations are

$$x \qquad \text{and} \qquad y - x.$$

Since

$$\gcd(x, y - x) = \gcd(x, y),$$

the quantity I_p is invariant under every move. In the final position all entries except M are 1, and hence

$$\nu_p(M) = I_p = \gcd\bigl(\nu_p(a_1), \ldots, \nu_p(a_{2026})\bigr),$$

where $a_1, \ldots, a_{2026}$ are the initial entries. This determines the exponent of every prime in M, so M is uniquely determined by the initial board. $\square$

2 Problem 2

Problem statement.

Let ABC be a triangle, and let M and N be the midpoints of sides AB and AC, respectively. Let K and L be chosen strictly inside triangles BMC and BNC, respectively, such that K lies strictly inside $\angle LBA$, L lies strictly inside $\angle ACK$, and

$$\angle KBA = \angle ACL, \qquad \angle LBK = \angle LNC, \qquad \angle LCK = \angle BMK.$$

Let O be the circumcenter of triangle AKL. Prove that $OM = ON$.

Core target. Establish a decisive relation equivalent to $OM = ON$. Equal powers with respect to (AKL) are one accepted route, not a required one.

Pts	Logical function	Criterion
1	**Extracting usable structure from the hypotheses**	For deriving correct, nontrivial consequences of the three angle conditions and the midpoint conditions that are used later in the proof. These may be cyclicities, directed-angle identities, trigonometric ratios, coordinate or complex equations, vector relations, or equivalent geometric structure. Merely restating the hypotheses earns no credit.
2	**Setting up a comparison of M and N**	For establishing a coherent framework that reduces $OM = ON$ to a concrete equality. Accepted frameworks include an equal-power configuration for the circumcircle (AKL), a formula for $OM^2 - ON^2$, or an equivalent synthetic, trigonometric, coordinate, vector, complex, or barycentric reduction. The auxiliary points of the reference solution are not mandatory. Award 1 point for a substantial correct setup that establishes one side or one essential relation but does not yet complete the reduction.
3	**Proving the decisive equality**	For rigorously proving $\mathrm{Pow}_{(AKL)}(M) = \mathrm{Pow}_{(AKL)}(N)$, or directly proving an equivalent decisive relation such as $OM^2 = ON^2$. Any valid method is accepted. In the reference route this is realized by locating appropriate secants and a correctly signed directed-product identity, but those objects are illustrative only. Award 1 point for a major correct identity that materially links the two sides, 2 points when the decisive equality is essentially obtained with a localized sign or justification gap, and all 3 points for a complete derivation.
1	**Completing the metric conclusion**	For correctly deducing $OM = ON$ from the established decisive equality, with signs, positivity, and any relevant nondegeneracy issues handled. If the preceding argument already proves $OM = ON$ directly, award this point for the complete final verification.
7	**Total**	

Coordination note. Directed angles and directed lengths may be used without separate casework. Under the usual notation, the powers are $\overline{MA}\,\overline{ME}$ and $\overline{NA}\,\overline{NF}$; writing $AM \cdot ME$ and $AN \cdot NF$ and calling them the powers without a sign convention is not fully correct. A non-directed solution must justify the relevant point order or signs. Equivalent equal-power constructions receive the same credit. An alternative proof may receive the corresponding points for establishing analogous cyclic or metric structure, identifying a suitable circle through A, K, L, proving equal powers of M, N, and completing the center–power argument; the named auxiliary points are not mandatory.

Reviewed reference solution

We use directed angles modulo 180° and directed products of collinear lengths. Put

$$X = BK \cap AC, \qquad Y = CL \cap AB.$$

The three angle conditions give, respectively, the cyclic quadrilaterals

$$B, Y, X, C; \qquad X, N, L, B; \qquad Y, M, K, C.$$

For example, the first condition is $\angle XBY = \angle XCY$; the other two are read in the same way from the defining collinearities. Because $MN \parallel BC$, the first cyclicity also gives that M, Y, X, N are cyclic. Denote the circles through X, N, L, B and Y, M, K, C by ω_B and ω_C, respectively.

Let the lines XK and XC meet ω_C again at K' and C'. Similarly, let YL and YB meet ω_B again at L' and B'. We first locate B' and C'. Taking powers of A with respect to ω_C and to the circle $(MYXN)$ gives

$$\overline{AC'}\,\overline{AC} = \overline{AY}\,\overline{AM} = \overline{AX}\,\overline{AN} = \overline{AX} \cdot \frac{\overline{AC}}{2}.$$

Thus $\overline{AC'} = \overline{AX}/2$, so C' is the midpoint of AX. In the same way, B' is the midpoint of AY.

Let E and F be the midpoints of BY and CX, respectively. We claim that

$$A, K, L, E, F, K', L'$$

lie on one circle. Indeed, power of X with respect to ω_C and the midpoint relations give

$$\overline{XK}\,\overline{XK'} = \overline{XC'}\,\overline{XC} = \overline{XA}\,\overline{XF},$$

so A, K, F, K' are cyclic. Also, taking power of B with respect to ω_C,

$$\overline{BK}\,\overline{BK'} = \overline{BM}\,\overline{BY} = \overline{BA}\,\overline{BE},$$

so A, E, K, K' are cyclic. Hence K and K' lie on the circumcircle of AEF. The symmetric argument with ω_B places L and L' on the same circle. In particular, it is the circumcircle of AKL; let its radius be ρ.

It remains to compare the powers of M and N. Since E is the midpoint of BY and M is the midpoint of BA, we have $\overline{ME} = \overline{AY}/2$. Similarly, $\overline{NF} = \overline{AX}/2$. Using the cyclic quadrilateral $MYXN$ and reversing the two directed segments that point toward A,

$$\overline{MA}\,\overline{ME} = -\frac{1}{2}\overline{AM}\,\overline{AY} = -\frac{1}{2}\overline{AN}\,\overline{AX} = \overline{NA}\,\overline{NF}.$$

These are precisely the powers of M and N with respect to the circumcircle of AKL. Since its center is O,

$$OM^2 - \rho^2 = ON^2 - \rho^2,$$

and therefore $OM = ON$. $\square$

3 Problem 3

Problem statement.

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick; all marked points are distinct. The stick is cut at all marked points. Afterwards, they take turns claiming any unclaimed piece, with Liu Bang going first. Each player seeks to maximize the total length of his own pieces. Determine the largest number c_n that Liu Bang can guarantee, regardless of Xiang Yu's play.

Core target. $c_n = \dfrac{2^n}{2^{n+1}-1}$.

Pts	Logical function	Criterion
1	**Finding the optimal value**	One point should be given for the correct answer $c_n = \frac{2^n}{2^{n+1}-1}$. This answer point may be awarded without a complete proof.
1	**Valuing the drafting phase**	For proving the value of the alternating drafting game on fixed piece lengths and correctly relating it to Liu Bang's share. One accepted formulation is that, for $z_1 \geq \cdots \geq z_m \geq 0$, taking a largest remaining piece is optimal and the score difference is $z_1 - z_2 + z_3 - \cdots + (-1)^{m+1} z_m$, so Liu Bang's share is $(1+G)/2$.
2	**Proving Xiang Yu's sharp upper bound**	For giving a legal response strategy, or an equivalent universal argument, that guarantees $G \leq 1/(2^{n+1}-1)$. The reference route pairs equal pieces and combines a cutting lemma with two close subset sums, but any sharp cancellation, combinatorial, or analytic proof is accepted. Award 1 point for a correct substantial lemma reducing the upper bound to a finite closeness or cancellation statement, and the second point for obtaining the stated sharp bound with all legality and edge cases resolved.
1	**Choosing an extremal division for Liu Bang**	For proposing the division in the ratio $1:2:4:\cdots:2^n$, or another construction proved to have the same extremal role, and correctly reducing the desired guarantee to the corresponding normalized lower bound.
2	**Proving Liu Bang's sharp lower bound**	For proving that the proposed division guarantees $G \geq 1/(2^{n+1}-1)$ and hence the claimed value of c_n, regardless of Xiang Yu's cuts. The even-index partial-sum estimate in the reference solution is one accepted route, not a required one. Award 1 point for a correct key inequality or structural lemma with substantial justification but an incomplete finish, and both points for the full lower bound.
7	**Total**	

Coordination note. The upper-bound strategy for Xiang Yu and the lower-bound strategy for Liu Bang are independent and should be scored independently. Coincident marks are not legal in the stated game. Zero-length pieces may be appended only as bookkeeping after the legal cuts, or an equivalent normalization must be justified. The cutting lemma must exclude $S = T = \varnothing$ (equivalently, require S and T to be distinct). The drafting-game value is a separately scored logical bridge: later bounds stated only in terms of G do not replace it unless the connection to optimal play is independently established.

Reviewed reference solution

The answer is

$$\boxed{c_n = \frac{2^n}{2^{n+1}-1}}.$$

For a sorted finite list $z_1 \ge z_2 \ge \cdots \ge z_m \ge 0$, the value of the drafting game in score difference for the player to move is

$$z_1 - z_2 + z_3 - \cdots + (-1)^{m+1} z_m. \tag{1}$$

This follows by induction on m: taking z_1 attains the displayed value, while if one takes z_j instead, the induction hypothesis for the remaining list shows that the loss relative to taking z_1 is twice a sum of nonnegative adjacent differences $z_1 - z_2, z_3 - z_4, \ldots$. Thus taking a largest remaining piece is optimal at every turn.

If Liu Bang makes fewer than n effective cuts, he creates at most n positive segments. Xiang Yu can bisect every one of them, using at most n marks, and force a score difference of 0. Thus, for the upper bound, we may assume Liu Bang makes exactly n effective cuts and creates $n+1$ positive original segments. If Xiang Yu uses fewer than n marks, pad the final list with zero-length pieces for bookkeeping; this does not change either score. We may therefore list $2n+1$ lengths in nonincreasing order:

$$x_1 \ge x_2 \ge \cdots \ge x_{2n+1} \ge 0.$$

By (1), Liu Bang receives $x_1 + x_3 + \cdots + x_{2n+1}$. Define the score difference

$$G = x_1 - x_2 + x_3 - x_4 + \cdots + x_{2n+1}.$$

Since the two players' total lengths sum to 1, Liu Bang's score is $(1+G)/2$. It is therefore enough to prove that the value of G is

$$\delta_n := \frac{1}{2^{n+1}-1}.$$

Xiang Yu can force $G \le \delta_n$. Suppose Liu Bang's n cuts produce $n+1$ original segments. For a set S of these segments, let $\Sigma(S)$ be their total length.

Lemma 1. *If S and T are disjoint, distinct sets of original segments, then Xiang Yu can make at most n cuts so that*

$$G \le \bigl|\Sigma(S) - \Sigma(T)\bigr|.$$

Proof. Assume $\Sigma(S) \ge \Sigma(T) > 0$. Concatenate the segments in S in any order and do the same for T, aligning the two concatenations at their left endpoints. On the common portion of length $\Sigma(T)$, Xiang Yu transfers every internal boundary from either concatenation to the other one. The resulting subpieces on the common portion occur in equal pairs.

Let $s = |S|$, $t = |T|$, and let $r = n + 1 - s - t$ be the number of original segments outside $S \cup T$. Suppose the common portion meets h segments of the S-concatenation. Creating the matching partition uses at most

$$(h-1) + (t-1) + 1 = h + t - 1$$

cuts; the final $+1$ is needed only when the common portion ends inside an S-segment. Xiang Yu bisects all r outside segments. In the overhang, he leaves its first component unpaired and bisects every later complete S-segment, using at most $s - h$ more cuts. Hence the total is at most

$$(h + t - 1) + r + (s - h) = s + t + r - 1 = n.$$

If the common portion ends at an existing boundary, or if the two totals are equal, even fewer cuts are needed.

After these cuts, every piece length occurs in a pair except possibly the first component of the overhang. Its length is at most $\Sigma(S) - \Sigma(T)$. In the alternating sum defining G, all equal pairs cancel, leaving at most that unpaired length. If $T = \varnothing$, Xiang Yu instead leaves one segment of S unpaired and bisects every other original segment; again at most n cuts are used, and the same bound follows. ☐

There are 2^{n+1} subset sums $\Sigma(S)$, all lying in $[0,1]$. After sorting them, two consecutive sums differ by at most

$$\frac{1}{2^{n+1}-1}.$$

Choose corresponding distinct subsets and delete their common elements. The resulting sets are disjoint and still distinct, and their subset-sum difference is unchanged. The lemma therefore gives $G \le \delta_n$.

Liu Bang can force $G \ge \delta_n$. He cuts the stick into lengths proportional to

$$1, 2, 4, \ldots, 2^n.$$

Scale temporarily so that these lengths are exactly $1, 2, \ldots, 2^n$; their total is $2^{n+1} - 1$. We prove that $G \ge 1$.

For $1 \le k \le n$, we claim

$$x_2 + x_4 + \cdots + x_{2k} \le 2^{n-1} + 2^{n-2} + \cdots + 2^{n-k}. \tag{2}$$

Assume that (2) first fails at k. Its validity for $k-1$ then implies $x_{2k} > 2^{n-k}$. Hence each of the first $2k$ pieces is longer than 2^{n-k}. Such pieces can come only from the k largest original segments

$$2^{n-k+1}, 2^{n-k+2}, \ldots, 2^n.$$

Because $x_{2i} \le x_{2i-1}$ for every i,

$$x_2 + x_4 + \cdots + x_{2k} \le \frac{x_1 + x_2 + \cdots + x_{2k}}{2} \le \frac{2^{n-k+1} + \cdots + 2^n}{2},$$

which is exactly the right-hand side of (2), a contradiction.

Taking $k = n$ in (2), Xiang Yu's total is at most $1 + 2 + \cdots + 2^{n-1} = 2^n - 1$. Liu Bang therefore receives at least 2^n, so $G \ge 1$ in the scaled game. Dividing by $2^{n+1} - 1$ gives $G \ge \delta_n$ in the unit-length game.

Thus the optimal score is

$$\frac{1 + \delta_n}{2} = \frac{1}{2}\left(1 + \frac{1}{2^{n+1} - 1}\right) = \frac{2^n}{2^{n+1} - 1}.$$

$\square$

4 Problem 4

Problem statement.

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$, known to both players. Initially, Shan-Yu makes a paper triangle T with measurements of his choice. They then repeat the following steps:

- If T has an angle measuring exactly θ, the game stops and Mulan wins.
- Otherwise, Mulan chooses a nonvertex point P on the perimeter of T and makes a straight cut from P to the opposite vertex, splitting T into two triangles.
- Shan-Yu discards one of the two triangles; the other becomes the new T.

For which real values of θ can Mulan guarantee victory after finitely many steps, no matter how Shan-Yu plays?

Core target. $\theta = \dfrac{180^\circ}{n}$ for integers $n \geq 2$.

Pts	Logical function	Criterion
1	**Finding the winning angles**	One point should be given for the correct classification $\theta = \frac{180^\circ}{n}$ for an integer $n \geq 2$.
1	**Establishing a finite winning-state mechanism**	For proving a lemma that lets Mulan force an angle θ from an appropriate state related to θ. The reference solution uses induction from an angle $k\theta$, but any equivalent finite-descent or forcing mechanism earns the point. A direct proof combining this mechanism with the next item may receive the combined 3 points.
2	**Proving Mulan can force a winning state**	For $\theta = 180^\circ/n$, construct and justify a legal strategy from an arbitrary initial triangle such that, no matter which child Shan-Yu keeps, Mulan reaches the winning-state mechanism and wins after finitely many moves. The altitude, right-triangle, and $k = \lfloor n/2 \rfloor$ construction is one full-credit method, not a mandatory route. Award 1 point for a substantial correct construction that covers one branch or all but a localized boundary case.
2	**Proving an obstruction in all remaining cases**	For θ not of the stated form, identifying and proving an invariant or obstruction under Shan-Yu's choice: after every legal cut, at least one child remains in a class of triangles that avoids an angle θ. The safe-angle invariant is one accepted realization, but any equivalent closed family or obstruction earns the same credit. Award 1 point for identifying a valid invariant and substantially checking its preservation, with one case or implication incomplete.
1	**Completing Shan-Yu's counterstrategy**	For giving a legal initial triangle in the invariant class and explaining how Shan-Yu retains an admissible child after every cut indefinitely, thereby proving that Mulan cannot force victory. A direct complete counterstrategy combining this with the previous item may receive the combined 3 points.
7	**Total**	

Coordination note. The altitude/right-triangle construction and the safe-angle invariant are one reference route, not mandatory checkpoints. Any complete finite winning strategy for $\theta = 180^\circ/n$ receives the three sufficiency points, and any valid invariant or counterstrategy for all remaining θ receives the three necessity points. In the displayed reference route, a triangle is safe only when all three angles are safe, and the appropriate angle identities must be used according to which new angle is a multiple of θ.

Reviewed reference solution

The winning angles are exactly

$$\boxed{\theta = \frac{180^\circ}{n} \quad \text{for an integer } n \geq 2.}$$

First suppose $\theta = 180^\circ/n$. We use the following observation: if the current triangle has an angle $k\theta$ for a positive integer k, then Mulan can force a win. This follows by induction on k. The case $k = 1$ is immediate. For $k > 1$, draw from that vertex the unique interior ray that splits the angle into θ and $(k-1)\theta$; it meets the opposite side at a nonvertex point. If Shan-Yu keeps the first child, the angle θ is already present; if he keeps the second, apply the induction hypothesis.

If the initial triangle does not already contain θ, Mulan first cuts along an altitude whose foot lies in the relative interior of the opposite side. Such an altitude always exists, for example from a vertex of a largest angle. Whichever child Shan-Yu keeps is a right triangle. Name it ABC so that

$$\angle A = 90^\circ, \qquad \angle B \leq 45^\circ.$$

For $n = 2$ this triangle already contains $\theta = 90^\circ$. Assume $n \geq 3$ and take $k = \lfloor n/2 \rfloor$. Since

$$\frac{n}{4} < \left\lfloor \frac{n}{2} \right\rfloor \leq \frac{n}{2},$$

we have

$$45^\circ < k\theta \leq 90^\circ.$$

Thus $0^\circ < k\theta - \angle B < 90^\circ$, so there is a point D in the interior of BC such that

$$\angle BAD = k\theta - \angle B.$$

Then

$$\angle ADB = 180^\circ - k\theta = (n-k)\theta, \qquad \angle ADC = k\theta.$$

Thus either child retained by Shan-Yu has an angle that is a positive integral multiple of θ, and the observation finishes the winning strategy.

Conversely, suppose θ is not of the form $180^\circ/n$. Call an angle *safe* if it is not an integral multiple of θ, and call a triangle safe if all three of its angles are safe. In particular, 180° is safe. We claim that a safe triangle cannot be cut into two triangles that are both unsafe.

Indeed, suppose D lies on BC and ABD is unsafe, while ABC is safe. Since $\angle B$ is safe, at least one of $\angle BAD$ and $\angle ADB$ must be an integral multiple of θ. If $\angle BAD$ is such a multiple, then

$$\angle CDA = \angle B + \angle BAD, \qquad \angle DAC = \angle A - \angle BAD$$

are both safe. If instead $\angle ADB$ is such a multiple, then

$$\angle CDA = 180^\circ - \angle ADB, \qquad \angle DAC = \angle ADB - \angle C$$

are both safe. In either case, adding or subtracting an integral multiple of θ cannot turn a safe angle into a multiple of θ. The remaining angle $\angle C$ was already safe, so ACD is safe. Therefore at least one child of every cut is safe.

Shan-Yu may start with an equilateral triangle. Its angles are safe, because $60^\circ = k\theta$ would imply $\theta = 180^\circ/(3k)$, contrary to the assumption. After every cut he keeps a safe child. No retained triangle ever has an angle θ, so Mulan cannot force a win. $\square$

5 Problem 5

Problem statement.

Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that, for every $x, y > 0$,

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{f(x) + y}{2} \geq \sqrt{xf(y)}.$$

Core target. $f(x) = x + c$ for a constant $c \geq 0$.

Pts	Logical function	Criterion
1	**Finding and checking the solution family**	For identifying the complete family $f(x) = x + c$ with $c \geq 0$ and verifying it in the original inequalities. The identity case $c = 0$ must be included.
2	**Deriving a rigid structural constraint**	For extracting from the two inequalities a strong relation that controls the iterates or displacement of f. In the reference route, substituting $(f(t), t)$ gives $f(f(t)) = 2f(t) - t$, whence $f^m(t) = t + m(f(t) - t)$ and $f(t) \geq t$. Any equivalent structural reduction is accepted. Award 1 point for the decisive relation alone and both points when its main orbit or sign consequences are rigorously derived.
2	**Showing the positive displacement is unique**	For proving that all positive values of $f(x) - x$ are equal, or establishing an equivalent rigidity statement that leaves at most one nonzero displacement. The square-root spacing argument on iterates is one accepted method, not a required one. Award 1 point for a correct decisive comparison or inequality with a substantial but incomplete equality argument.
2	**Ruling out mixed behavior and concluding**	For proving that fixed points and points shifted by the common positive displacement cannot coexist unless all points are fixed, and hence that $f(x) - x$ is constant on $\mathbb{R}_{>0}$. The separation-and-propagation argument in the reference solution is illustrative only. Award 1 point for a valid local separation or propagation lemma without the global conclusion, and both points for completing the classification.
7	**Total**	

Coordination note. Evan Chen's 17 July notes display $c > 0$, but the identity function also satisfies the inequalities. A claimed complete answer omitting $c = 0$ is incomplete. The reviewed proof correctly gives $c \geq 0$. Alternative proofs receive the same functional credit for establishing a common nonzero displacement, ruling out coexistence of fixed and displaced points, and propagating the displacement globally.

Reviewed reference solution

The complete family is

$$\boxed{f(x) = x + c \quad \text{for some constant } c \geq 0.}$$

Every such function works, because the required chain becomes

$$\sqrt{\frac{x^2 + (y + c)^2}{2}} \geq \frac{x + y + c}{2} \geq \sqrt{x(y + c)},$$

which is the quadratic-mean–arithmetic-mean–geometric-mean inequality for the positive numbers x and $y+c$.

We prove that there are no other functions. Let $P(x,y)$ denote the left inequality and $Q(x,y)$ the right inequality. Substituting $(x,y)=(f(t),t)$, the two outer expressions are both $f(t)$, so equality is forced in the middle:

$$\frac{f(f(t))+t}{2}=f(t). \tag{1}$$

Thus $f(f(t))=2f(t)-t$. If $d(t):=f(t)-t$, then (1) says $d(f(t))=d(t)$, and induction gives

$$f^m(t)=t+md(t), \qquad m=0,1,2,\dots. \tag{2}$$

All iterates are positive, so $d(t)\geq 0$.

We next show that all positive values of $d(t)$ are equal. Choose $x,y>0$ with $d(x)=A>0$ and $d(y)=B>0$. From (2),

$$f^m(x)=x+mA, \qquad f^n(y)=y+nB.$$

Apply Q to $f^m(x)$ and $f^n(y)$, and put

$$U=x+mA, \qquad V=y+(n+1)B.$$

The resulting inequality is

$$\frac{U+V+A-B}{2}\geq\sqrt{UV},$$

which is equivalent to

$$\left(\sqrt{x+mA}-\sqrt{y+(n+1)B}\right)^2\geq B-A. \tag{3}$$

This holds for all nonnegative integers m,n. If $A<B$, set $\varepsilon=\sqrt{B-A}$ and write

$$s_m=\sqrt{x+mA}, \qquad t_n=\sqrt{y+(n+1)B}.$$

The sequence (t_n) is unbounded and $t_{n+1}-t_n\to 0$. Choose N so that $t_{n+1}-t_n<\varepsilon$ for all $n\geq N$, and then choose m with $s_m>t_N$. For the index $n\geq N$ satisfying $t_n\leq s_m<t_{n+1}$, at least one of $|s_m-t_n|$ and $|s_m-t_{n+1}|$ is less than ε, contradicting (3). Hence $A\geq B$. By symmetry, $B\geq A$, so $A=B$.

Therefore either $d(t)=0$ for every t, giving the identity function, or there is a single constant $D>0$ such that

$$f(t)\in\{t,t+D\} \qquad (t>0). \tag{4}$$

Assume the second case. If there is no fixed point, then $f(t)=t+D$ for all t and we are done. Otherwise choose a point x with $f(x)=x+D$ and a fixed point y with $f(y)=y$. Necessarily $x\neq y$. If $x>y$, then $P(x,y)$ yields

$$\frac{x+y+D}{2}\leq\sqrt{\frac{x^2+y^2}{2}}<x,$$

so $x>y+D$. If $x<y$, then $Q(y,x)$ yields

$$\sqrt{y(x+D)}\leq\frac{x+y}{2}<y,$$

so $x<y-D$. In either case,

$$|x-y|>D. \tag{5}$$

Consequently, whenever $f(x)=x+D$, every positive t with $|t-x|\leq D$ must also satisfy $f(t)=t+D$. Given any $t>0$, subdivide the interval joining x to t into finitely many steps of length at most D. Applying the preceding implication along this chain shows successively that every subdivision point, and finally t, satisfies $f(t)=t+D$. Hence $f(t)=t+D$ for all $t>0$.

Combining this with the identity case gives $f(x)=x+c$ for exactly the constants $c\geq 0$. $\square$

6 Problem 6

Problem statement.

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that, for every positive integer n, the number a_{n+1} is the smallest positive integer greater than a_n such that

$$\gcd(a_{n+1}, a_i) > 1 \qquad \text{for every } i = 1, 2, \dots, n.$$

Prove that there exist positive integers T and L such that

$$a_{n+T} = a_n + L$$

for every positive integer n.

Core target. Show that sequence membership is governed by finitely many periodic divisibility features, then derive the exact translation law $a_{n+T} = a_n + L$.

Pts	Logical function	Criterion
1	**Establishing an occurrence criterion**	For characterizing which integers $x \geq a_1$ occur in the sequence in terms of a nonredundant family of earlier gcd constraints, or an equivalent criterion that can be checked independently of the order of intermediate selections. The radical-minimal relation in the reference solution is one accepted construction, not a mandatory one.
3	**Proving that only finitely many prime features matter**	For proving that the occurrence criterion is controlled by a finite set of primes or finite divisibility features. In the reference route this is the lemma excluding primes $p > a_1^2$ from radical-minimal terms. Award the 3 points by logical function: 1 for a valid finite-bound/redundancy setup, 1 for the central witness or inductive step showing that a new large prime cannot be essential, and 1 for completing the exclusion and finite-control conclusion. Any equivalent finite-prime argument receives the same credit.
2	**Proving periodicity of sequence membership**	For showing that, among integers $a \geq a_1$, membership in the sequence depends only on finitely many divisibility signatures and therefore only on the residue class modulo some fixed positive integer L. Award 1 point for proving one direction of the signature characterization or showing periodic dependence without the exact converse, and both points for a complete if-and-only-if periodic membership criterion.
1	**Converting periodic membership to the translation law**	For ordering the recurring residue classes, choosing a positive integer T, and rigorously deducing $a_{n+T} = a_n + L$ for every $n \geq 1$. Eventual periodicity alone does not earn this final point unless it is strengthened to the required all-n translation law.
7	**Total**	

Coordination note. In the large-prime lemma, if k is the least nonnegative integer with $q^k c \geq a_1$, the estimate $q^k c < qa_1$ is valid only for $k \geq 1$. The case $k = 0$ must be handled separately by $b = c < a_n$, or replaced by an equivalent multiplicative-interval argument. The membership criterion is for $x \geq a_1$, not arbitrary positive x. Route-neutral scoring is: reduce the infinitely many gcd constraints to a determining family; prove that only finitely many primes occur in that family; classify membership by finitely many divisibility signatures; and deduce the translation law.

Reviewed reference solution

For a positive integer u, let $\operatorname{rad}(u)$ be the product of the distinct primes dividing u. Define a relation among terms of the sequence by

$$a_m \prec a_n \quad \iff \quad m < n \text{ and } \operatorname{rad}(a_m) \mid \operatorname{rad}(a_n).$$

If $a_m \prec a_n$, then the condition of sharing a prime with a_n is redundant: any integer sharing a prime with a_m automatically shares one with a_n. Every nonminimal term has, after finitely many descents in the index, a $\prec$-minimal predecessor whose radical divides its own. Consequently, an integer $x \geq a_1$ belongs to the sequence if and only if it has gcd greater than 1 with every $\prec$-minimal term smaller than x.

Necessity is immediate. Conversely, this condition makes x share a prime with every term of the sequence smaller than x. Hence, while the current sequence term is below x, the integer x is an admissible candidate for the next term. The sequence cannot jump past x, and only finitely many integers lie below x, so it must eventually select x.

Call a prime *large* if it exceeds a_1^2. The key point is that no $\prec$-minimal term is divisible by a large prime.

Lemma 2. *If a_n is divisible by a prime $p > a_1^2$, then a_n is not $\prec$-minimal.*

Proof. We argue by induction on n. Write $a_n = pc$. The case $n = 1$ is impossible, since every prime divisor of a_1 is at most a_1. For $n > 1$, choose a prime q dividing $\gcd(a_1, a_n)$; such a prime exists by the definition of the sequence. Let k be the least nonnegative integer for which $q^k c \geq a_1$, and set $b = q^k c$. If $k = 0$, then $b = c < a_n$. If $k \geq 1$, minimality of k gives $q^{k-1}c < a_1$, and hence

$$b = q^k c < qa_1 \leq a_1^2 < p \leq pc = a_n.$$

Thus in every case

$$b \in [a_1, a_n).$$

Consider any $\prec$-minimal earlier term $a_i < a_n$. By the induction hypothesis, $p \nmid a_i$. Since $\gcd(a_n, a_i) > 1$, some common prime divisor of a_n and a_i is different from p and therefore divides c. Hence $\gcd(b, a_i) > 1$.

Thus b satisfies all conditions imposed by the $\prec$-minimal predecessors, so the criterion above shows that b occurs in the sequence. Moreover every prime divisor of $b = q^k c$ divides $a_n = pc$, because $q \mid a_n$. Hence $\operatorname{rad}(b) \mid \operatorname{rad}(a_n)$ and $b < a_n$, so $b \prec a_n$. Therefore a_n is not minimal. □

Let $\mathcal{P}$ be the finite set of primes at most a_1^2, and define

$$S(a) = \{p \in \mathcal{P} : p \mid a\}, \qquad \mathcal{F} = \{S(a_n) : n \geq 1\}.$$

We claim that the sequence consists exactly of the integers $a \geq a_1$ satisfying $S(a) \in \mathcal{F}$.

One direction is immediate. Conversely, suppose $S(a) = S(a_n)$ for some n. Any two terms of the sequence have gcd greater than 1, because the later term was chosen to share a prime with the earlier one. Let a_i be a $\prec$-minimal term. By the lemma, all prime divisors of a_i lie in $\mathcal{P}$. Choose a common prime divisor of a_i and a_n; it lies in $S(a_n) = S(a)$, so it also divides a. Thus a shares a prime with every minimal term, and the criterion above shows that a occurs in the sequence.

Now set

$$L = \prod_{p \in \mathcal{P}} p.$$

The set $S(a)$ depends only on the residue class of a modulo L. Hence membership in the sequence, among integers $a \geq a_1$, is periodic modulo L. Let

$$\mathcal{R} = \{r \in \mathbb{Z}/L\mathbb{Z} : S(a) \in \mathcal{F} \text{ for } a \equiv r \pmod{L}\}, \qquad T = |\mathcal{R}|.$$

In the interval $(a_n, a_n + L]$ there is exactly one representative of each class in $\mathcal{R}$, and no representative of any other class belongs to the sequence. Thus this interval contains exactly the next T terms, with $a_n + L$ as the last one. Consequently

$$a_{n+T} = a_n + L$$

for every $n \geq 1$. □